- `geo_shape` *query*—Finds addresses represented as shapes within a shape constructed by a set of coordinates; for example, agricultural farms contained in a green belt, where the farms and the green belt are represented by individual shapes as well as the green belt

We discuss queries for both geopoints and geoshapes in detail in the following sections. Let's begin with queries executed on `geo_point` fields.

12.4 The geo_bounding_box query

When we search for a list of addresses, we can use an area of interest. This area can be represented by a circle of a certain radius, an area enclosed by a shape such as a rectangle, or a polygon from a central point (a landmark).

Elasticsearch provides a `geo_bounding_box` query that lets us search for locations inside these areas. For example, as figure 12.4 shows, we can construct a rectangle using latitude and longitude coordinates and search to see if our address exists in that area.



Figure 12.4 Georectangle from latitude and longitude coordinates

The `top_left` and `bottom_right` fields are the coordinates of latitude and longitude that make up our georectangle. Once we define a georectangle, we can check if the points of interest (the Imperial College London, for example) are present inside this rectangle.

Before we discuss the `geo_bounding_box` query in detail, let's write the query; then we can dissect it. The following query searches for all documents (locations) that fit in the rectangle constructed with `top_left` and `bottom_right` coordinates.